Sparse Rank Witnesses for Dyadic-Free Cubic Graphs

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Abstract

For an integer $R \geq 3$, let G be a connected simple cubic graph on $n > 2(R-1)$ vertices. We prove that G contains either a cycle of a specified power-of-two length or a connected subgraph of cycle rank $R+1$ with $O(R^3(\log\log n + \log R))$ edges. Consequently, every sufficiently large connected dyadic-free cubic graph contains a connected subgraph with fewer than $\log_2 n$ edges and cycle rank greater than $\frac{1}{16}(\log_2 n/\log_2(\log_2 n+2))^{1/3}$. The proof places all short cycles in bounded-rank subgraphs with pairwise disjoint neighbourhoods, each consisting of its subgraph with trees attached. Resolvent estimates on these subgraphs then bound the nonsimple closed nonbacktracking walks in a trace calculation. Pruning and suppressing the resulting rank witness gives a cubic multigraph whose positive integral edge lengths have total less than $\log_2 n$.

1 Introduction and main result

A *dyadic cycle* is a simple cycle of length 2^k for an integer $k \geq 2$. A graph is *dyadic-free* if it contains no such cycle. The Erdős–Gyárfás conjecture asserts that every nonempty finite simple graph of minimum degree at least three contains a dyadic cycle [Erd97]. We study the cubic case, where every vertex has degree three. Heckman and Krakovski prove the conjecture for 3-connected cubic planar graphs [HK13]. In a preprint, Ducoffe and Dumitru study structural restrictions on minimal counterexamples and report computational verification for cubic graphs of order at most 48 [DD26].

All graphs are finite. They are simple unless explicitly described as multigraphs. A subcubic graph has maximum degree at most three. For a graph J, write

$$r(J) = |E(J)| - |V(J)| + \kappa(J),$$

where $\kappa(J)$ is the number of connected components. This is the *cycle rank*; equivalently, it is the dimension of the cycle space over $\mathbb{F}_2$. Indeed, choose a spanning forest. Each edge outside it creates one fundamental cycle, and these cycles form a basis: they are independent on the nonforest edges, and subtracting them from any cycle-space vector leaves an even-degree subgraph of a forest, which is empty. There are $|E(J)|-|V(J)|+\kappa(J)$ nonforest edges. For a connected graph, $r(J) = |E(J)| - |V(J)| + 1$. The order of a graph is its number of vertices.

Distances are measured in the ambient graph G. For a nonempty subgraph K and an integer $h \geq 0$, set

$$d_G(v, K) = \min_{x \in V(K)} d_G(v, x), \qquad N_h(K) = G[\{v : d_G(v, K) \leq h\}].$$

Thus $B_G(v, h) = N_h(\{v\})$. These neighbourhoods are induced; other subgraphs may be arbitrary. For nonempty subgraphs K, K', write $d_G(K, K') = \min_{v \in V(K)} d_G(v, K')$. Write $|C| = |E(C)| = |V(C)|$ for the length of a simple cycle.

Vectors are real unless stated otherwise. For $x \in \mathbb{R}^{V(G)}$ and $J \subseteq G$,

$$\|x_J\|^2 = \sum_{v \in V(J)} x_v^2, \qquad \|M\| = \sup_{\|x\|=1} \|Mx\|.$$

All logarithms are base two except ln. For $x \geq 1$, set

$$\operatorname{pow}_2(x) = 2^{\lceil \log_2 x \rceil}, \qquad x \leq \operatorname{pow}_2(x) < 2x.$$

Theorem 1 (Sparse-rank dichotomy) *Let $R \geq 3$ be an integer, and let G be a connected simple cubic graph of order $n > 2(R-1)$. Define*

$$t = \log_2 n, \qquad u = \log_2(t+2), \qquad c = 16(R-1)^2,$$

$$h = \left\lceil 4c\big(u + \log_2(u+2) + 2\log_2(c+1) + 10\big) \right\rceil,$$

$$L = 2h + 1, \qquad W = (4R-1)L, \qquad q = \operatorname{pow}_2(2ht).$$

Then G contains either a simple cycle of length q or a connected subgraph J such that

$$r(J) = R + 1, \qquad |E(J)| \leq W.$$

In particular,

$$W = O\big(R^3(\log\log n + \log R)\big),$$

with an absolute implied constant.

Assuming the rank alternative fails, Lemma 8 places all cycles of length at most L in bounded-rank cores with disjoint tree neighbourhoods. Lemmas 13 and 15 bound eigenvector mass on the cores and the number of local walks. These estimates give $D_q < T_q$, where T_q counts closed nonbacktracking walks and D_q counts the nonsimple ones. The identity $T_q - D_q = 2qc_q$ then gives a simple q-cycle.

Spectral background. The determinant reduction is the Ihara–Bass formula [Bas92]; see also Kempton [Kem16, Section 2.4] for a directed-edge proof. The walk recurrence and its Chebyshev representation for regular graphs appear in Alon, Benjamini, Lubetzky, and Sodin [ABLS07, Section 2]. Rangarajan [Ran18, Section 2.2] gives the corresponding generating function and distinguishes rooted returns from cyclically nonbacktracking walks. We derive these identities in Section 2. Proposition 9 states a quantitative cycle criterion under explicit local rank and tree-neighbourhood hypotheses. Its proof in Sections 5–8 includes the core resolvent, eigenvector-mass estimates, and trace comparison. Section 9 applies the criterion to the subgraphs constructed in Section 3.

2 Closed nonbacktracking walks

2.1 Walks, roots, and cyclic closure

For a graph H, let

$$\vec{E}(H) = \{(v, w) : \{v, w\} \in E(H)\}.$$

For $e = (v, w)$, set $\mathrm{tail}(e) = v$, $\mathrm{head}(e) = w$, and $\bar{e} = (w, v)$. The nonbacktracking matrix B_H, indexed by $\vec{E}(H)$, is defined by

$$(B_H)_{ef} = \mathbf{1}_{\{\mathrm{head}(e)=\mathrm{tail}(f),\ f\neq\bar{e}\}}.$$

Write $B = B_G$. For $j \geq 1$ and $v, w \in V(H)$, let $\mathcal{W}_j^H(v, w)$ consist of the sequences $(v_0, \ldots, v_j)$ with

$$v_0 = v, \qquad v_j = w, \qquad \{v_{k-1}, v_k\} \in E(H) \quad (1 \leq k \leq j),$$
$$v_{k-1} \neq v_{k+1} \quad (1 \leq k < j).$$

These walks are *internally nonbacktracking*. At length zero, $\mathcal{W}_0^H(v, w)$ has one element when $v = w$ and is empty otherwise.

For a positive integer q, define the cyclically nonbacktracking walk set

$$\mathcal{T}_q = \{(e_0, \dots, e_{q-1}) \in \vec{E}(G)^q : B_{e_k e_{k+1}} = 1 \ (0 \le k < q)\},$$

where $e_q = e_0$. Matrix multiplication gives

$$T_q := |\mathcal{T}_q| = \sum_{e_0,\dots,e_{q-1}} \prod_{k=0}^{q-1} B_{e_k e_{k+1}} = \operatorname{tr}(B^q). \tag{1}$$

The tuple retains its initial directed edge. Put $v_k = \operatorname{tail}(e_k)$ and

$$\mathcal{D}_q = \{(e_0, \dots, e_{q-1}) \in \mathcal{T}_q : |\{v_0, \dots, v_{q-1}\}| < q\}, \qquad D_q = |\mathcal{D}_q|.$$

Let c_q be the number of undirected simple cycles of length q. Each element of $\mathcal{T}_q \setminus \mathcal{D}_q$ specifies a simple q-cycle, an orientation, and an initial edge. Conversely, each such choice gives one tuple. Thus

$$T_q - D_q = 2qc_q. \tag{2}$$

The restriction $B_{e_{q-1}e_0} = 1$ enforces nonbacktracking at the root; $\mathcal{W}_q^G(v, v)$ imposes only the internal restrictions. These two counting conventions are treated explicitly in [Ran18, Sections 2.1–2.2].

2.2 The determinant identity and the trace spectrum

Lemma 2 (Ihara–Bass identity) *Let A and D be the adjacency and degree matrices of a finite simple graph of order n and size m, and let B be its directed-edge nonbacktracking matrix. Then*

$$\det(I - zB) = (1 - z^2)^{m-n} \det\big(I - zA + z^2(D - I)\big).$$

The identity is interpreted after cancellation when $m < n$. In a cubic graph, the eigenvalues of B, counted with algebraic multiplicity, consist of $n/2$ copies of each of 1 and -1, together with the two roots of

$$\mu^2 - \lambda_i \mu + 2 = 0$$

for each adjacency eigenvalue λ_i.

Proof. Let $U, V \in \mathbb{R}^{n \times 2m}$ and $J \in \mathbb{R}^{2m \times 2m}$ be given by

$$U_{ve} = \mathbf{1}_{\{v = \operatorname{tail}(e)\}}, \qquad V_{ve} = \mathbf{1}_{\{v = \operatorname{head}(e)\}}, \qquad J_{ef} = \mathbf{1}_{\{f = \bar{e}\}}.$$

The definitions give

$$B = V^\top U - J, \quad UV^\top = A, \quad UJ = V, \quad UJV^\top = VV^\top = D, \quad J^2 = I.$$

For $z^2 \neq 1$, we have $(I+zJ)^{-1} = (I-zJ)/(1-z^2)$ and $\det(I+zJ) = (1-z^2)^m$, since J has one reversal block $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ per edge. For rectangular matrices X, Y of compatible sizes, $\det(I - XY) = \det(I - YX)$: eliminating either off-diagonal block of $\left(\begin{smallmatrix} I & X \\ Y & I \end{smallmatrix}\right)$ gives the two determinants. Factoring $I - zB$ and applying this identity gives

$$\begin{aligned} \det(I - zB) &= \det(I + zJ)\det\big(I - zU(I + zJ)^{-1}V^\top\big) \\ &= (1 - z^2)^m \det\left(I - \frac{z(A - zD)}{1 - z^2}\right) \\ &= (1 - z^2)^{m-n} \det\big(I - zA + z^2(D - I)\big). \end{aligned}$$

After clearing denominators, equality is a polynomial identity and therefore also holds at $z = \pm 1$.

For a cubic graph, $D = 3I$ and $m = 3n/2$. Diagonalizing the real symmetric matrix A yields

$$\det(I - zB) = (1 - z^2)^{n/2} \prod_{i=1}^{n} (1 - \lambda_i z + 2z^2),$$

Equivalently, as a polynomial identity in x,

$$\det(xI - B) = (x^2 - 1)^{n/2} \prod_{i=1}^{n} (x^2 - \lambda_i x + 2). \tag{3}$$

This proves the multiplicity assertion. The identity $\operatorname{tr}(B^q) = \sum_{\mu \in \operatorname{spec}(B)} \mu^q$ follows by upper-triangularizing B over $\mathbb{C}$, with eigenvalues repeated according to algebraic multiplicity. Taking a power raises each diagonal entry to that power. □

Every adjacency eigenvalue of a cubic graph lies in $[-3, 3]$: for every real vector x,

$$|x^\top A x| \leq \sum_{uv \in E(G)} (x_u^2 + x_v^2) = 3\|x\|^2.$$

For $|\lambda_i| > 2\sqrt{2}$, set

$$\beta_i = \frac{|\lambda_i| + \sqrt{\lambda_i^2 - 8}}{2}.$$

Then $\sqrt{2} < \beta_i \le 2$ and $|\lambda_i| = \beta_i + 2/\beta_i$. The two roots in (3) are $\mathrm{sgn}(\lambda_i)\beta_i$ and $\mathrm{sgn}(\lambda_i)2/\beta_i$. For $|\lambda_i| \le 2\sqrt{2}$, define $\theta_i = \arccos(\lambda_i/(2\sqrt{2})) \in [0, \pi]$; the roots are $\sqrt{2}e^{\pm i\theta_i}$, including multiplicities at the endpoints. Consequently, for every positive even integer q,

$$\begin{aligned} T_q = n + \sum_{|\lambda_i| > 2\sqrt{2}} \left(\beta_i^q + (2/\beta_i)^q\right) \\ + 2\, 2^{q/2} \sum_{|\lambda_i| \le 2\sqrt{2}} \cos(q\theta_i). \end{aligned} \tag{4}$$

For $\sqrt{2} < \alpha < 2$, write

$$\mathcal{I}_{>\alpha} = \{i : |\lambda_i| > 2\sqrt{2},\ \beta_i > \alpha\}.$$

Every sum indexed by $\beta_i > \alpha$ is over this set. Indices corresponding to repeated adjacency eigenvalues are retained.

Lemma 3 (Trace bounds) *For a nonempty simple cubic graph and a positive even integer q,*

$$T_q \ge 2^q - 2n\, 2^{q/2} = (1 - \eta_q)2^q, \qquad \eta_q = 2n\, 2^{-q/2}.$$

For $\sqrt{2} < \alpha < 2$, put $S_q(\alpha) = \sum_{\beta_i > \alpha} \beta_i^q$. If $\eta_q < 1$, then

$$S_q(\alpha) \le \frac{T_q}{1 - \eta_q}.$$

Proof. The adjacency eigenvalue 3 contributes $2^q + 1$ to the trace. For $|\lambda_i| \le 2\sqrt{2}$, both roots have modulus $\sqrt{2}$, so their combined contribution is at least $-2 \cdot 2^{q/2}$. The real-root pairs and the exceptional roots ± 1 have nonnegative even powers. These observations give both

$$T_q \ge 2^q - 2n\, 2^{q/2} \quad \text{and} \quad T_q \ge S_q(\alpha) - 2n\, 2^{q/2}.$$

When $\eta_q < 1$, the first inequality implies $2^q \le T_q/(1 - \eta_q)$. Substitution into the second gives

$$S_q(\alpha) \le T_q + \eta_q 2^q \le T_q + \frac{\eta_q T_q}{1 - \eta_q} = \frac{T_q}{1 - \eta_q}.$$

□

2.3 Walk polynomials and uniform spectral bounds

The following recurrence is the cubic specialization of [ABLS07, Section 2, equations (10)–(12)].

Lemma 4 (Walk polynomials) *In a cubic graph, for all $v, w \in V(G)$ and integers $j \geq 0$,*

$$(P_j(A))_{vw} = |\mathcal{W}_j^G(v, w)|,$$

where

$$\begin{aligned} &P_0(x) = 1, && P_1(x) = x, && P_2(x) = x^2 - 3, \\ &P_j(x) = xP_{j-1}(x) - 2P_{j-2}(x) && && (j \geq 3). \end{aligned}$$

For $\sqrt{2} < \beta \leq 2$, $\lambda = \beta + 2/\beta$, and $j \geq 1$,

$$P_j(\lambda) = \frac{\beta^2 - 1}{\beta^2 - 2}\beta^j - \frac{4 - \beta^2}{2(\beta^2 - 2)}(2/\beta)^j. \tag{5}$$

In particular, $|P_j(\lambda)| \geq \frac{3}{2}\beta^j$ when $|\lambda| = \beta + 2/\beta$. For $\sqrt{2} < \gamma < 2$, define $C_\gamma = (\gamma^2 - 1)/(\gamma^2 - 2)$. Then, for $j \geq 1$,

$$\begin{aligned} &|P_j(\lambda)| \leq C_\gamma \gamma^j && \text{if } |\lambda| \leq \gamma + 2/\gamma, \\ &|P_j(\lambda)| \leq C_\gamma \beta^j && \text{if } |\lambda| = \beta + 2/\beta, \quad \gamma < \beta \leq 2. \end{aligned}$$

Proof. For length two, the three immediate returns at each vertex must be removed from A^2. For $j \geq 3$, prepend an edge to a nonbacktracking walk of length $j - 1$. Each forbidden reversal is determined by a nonbacktracking walk of length $j - 2$ and one of the two incident edges other than its first edge. This proves the recurrence.

For $\lambda = \beta + 2/\beta$, the characteristic roots of the recurrence are β and $2/\beta$. Write $P_j(\lambda) = A_\beta \beta^j - B_\beta (2/\beta)^j$ for $j \geq 1$. The initial values give

$$A_\beta \beta - B_\beta (2/\beta) = \lambda, \qquad A_\beta \beta^2 - B_\beta (2/\beta)^2 = \lambda^2 - 3.$$

Solving yields

$$A_\beta = \frac{\beta^2 - 1}{\beta^2 - 2}, \qquad B_\beta = \frac{4 - \beta^2}{2(\beta^2 - 2)}, \qquad A_\beta - B_\beta = \frac{3}{2}.$$

This proves (5). Since $B_\beta \geq 0$ and $2/\beta \leq \beta$,

$$\frac{3}{2}\beta^j \leq P_j(\lambda) \leq A_\beta \beta^j.$$

Moreover, $A_\beta = 1 + 1/(\beta^2 - 2)$ decreases with β, so $A_\beta \le C_\gamma$ when $\beta > \gamma$. Induction in the recurrence gives $P_j(-x) = (-1)^j P_j(x)$, which proves the stated bounds on the negative real-root interval as well.

For the remaining spectral interval, define the Chebyshev polynomials by $U_{-1}(x) = 0$, $U_0(x) = 1$, $U_1(x) = 2x$, and $U_j(x) = 2xU_{j-1}(x) - U_{j-2}(x)$ for $j \ge 2$. Matching the initial values at $j = 1, 2$ and the recurrence gives

$$P_j(\lambda) = 2^{j/2}\left[U_j\left(\frac{\lambda}{2\sqrt{2}}\right) - \frac{1}{2}U_{j-2}\left(\frac{\lambda}{2\sqrt{2}}\right)\right] \quad (j \ge 1). \tag{6}$$

For $j \ge 0$, the finite sum

$$U_j(\cos\theta) = \sum_{k=0}^{j} e^{i(j-2k)\theta}$$

satisfies the same recurrence and initial values. It is valid for real or purely imaginary θ. Hence, for $j \ge 1$,

$$P_j(2\sqrt{2}\cos\theta) = 2^{j/2}\left(e^{ij\theta} + e^{-ij\theta} + \frac{1}{2}\sum_{k=1}^{j-1} e^{i(j-2k)\theta}\right). \tag{7}$$

The inner sum is empty at $j = 1$. For real θ, the triangle inequality gives

$$|P_j(2\sqrt{2}\cos\theta)| \le 2^{j/2}\left(2 + \frac{j-1}{2}\right) = P_j(2\sqrt{2}).$$

For $\theta = iy$, pair opposite exponents in (7). Each pair is a positive multiple of $\cosh(ky)$ for some integer $k \ge 0$, so $y \mapsto P_j(2\sqrt{2}\cosh y)$ is nondecreasing on $[0, \infty)$. Since $\gamma + 2/\gamma = 2\sqrt{2}\cosh(\ln(\gamma/\sqrt{2}))$,

$$|\lambda| \le \gamma + 2/\gamma \implies |P_j(\lambda)| \le P_j(\gamma + 2/\gamma) \le C_\gamma \gamma^j.$$

Parity supplies the implication for negative λ. □

Write

$$f_j(v) = (P_j(A))_{vv} = |\mathcal{W}_j^G(v, v)|,$$
$$s_r(v) = 2\,|\{C : C \text{ is a simple } r\text{-cycle},\ v \in V(C)\}|.$$

Thus $f_j(v)$ permits a reversal between the last and first edges, and $s_r(v)$ counts oriented simple r-cycles rooted at v. In particular,

$$0 \le s_r(v) \le f_r(v), \qquad \sum_v s_r(v) = 2rc_r.$$

2.4 Marking a repeated vertex

Lemma 5 (Marking a repeated vertex) *For a simple cubic graph and $q \geq 3$,*

$$D_q \leq q \sum_{r=3}^{\lfloor q/2 \rfloor} \sum_{v} s_r(v) f_{q-r}(v).$$

Moreover, $s_r(v) \leq f_r(v)$. If the girth is g, then $D_q = 0$ for $q < 2g$.

Proof. Consider a nonsimple walk counted by T_q. Among all positive cyclic intervals joining two distinct positions occupied by the same vertex, choose one of minimum length, breaking ties by the order of the positions. Either of the two intervals between any repeated pair is eligible, so the chosen length r is at most $q/2$. No internal vertex of the chosen interval repeats, since that would give a shorter interval. Its length cannot be one in a simple graph, or two in a nonbacktracking walk. It is therefore a simple cycle of length at least three.

The complementary interval is an internally nonbacktracking return of length $q-r$ at the marked vertex. If the mark is at position $k \in \{0, \ldots, q-1\}$, concatenate the oriented cycle with the complementary return and rotate its edge sequence so that the mark occupies position k. This reconstructs the original ordered tuple. The deterministic choice of the shortest interval therefore defines an injection into the disjoint union

$$\bigsqcup_{r=3}^{\lfloor q/2 \rfloor} \bigsqcup_{v \in V(G)} \big(\{0, \ldots, q-1\} \times \mathcal{S}_r(v) \times \mathcal{W}^G_{q-r}(v, v) \big),$$

where $\mathcal{S}_r(v)$ is the set of oriented simple r-cycles rooted at v and $|\mathcal{S}_r(v)| = s_r(v)$. The target allows all transition choices between the two pieces. Taking cardinalities proves the inequality. A simple rooted cycle is one of the returns counted by $f_r(v)$, giving $s_r(v) \leq f_r(v)$.

Any repeated vertex also divides a cyclic nonbacktracking walk into two nonempty internally nonbacktracking closed walks. Each contains a cycle: a nonempty closed walk on a tree has an immediate reversal at an occurrence of a vertex farthest from its starting vertex. Thus each piece has length at least g, and the whole walk has length at least $2g$. □

3 Short cycles and disjoint tree neighbourhoods

Fix the parameters of Theorem 1. Throughout this section, assume that the family

$$\{J \subseteq G : J \text{ is connected},\ r(J) = R+1,\ |E(J)| \le W\} = \varnothing. \tag{8}$$

This also excludes connected subgraphs of rank at least $R+1$ with at most W edges. Indeed, if T is a spanning tree of such a subgraph J and $F \subseteq E(J) \setminus E(T)$ has $R+1$ edges, then

$$J' = (V(J), E(T) \cup F), \qquad r(J') = |F| = R+1, \qquad |E(J')| \le |E(J)| \le W.$$

Lemma 6 (Rank of a ball) *Every induced ball $B_G(v,h)$ has cycle rank at most R.*

Proof. Otherwise take a breadth-first spanning tree of the ball and select $R+1$ edges outside the tree. Keep these edges and the tree paths from their endpoints to v. The tree paths form a connected tree, so adding the selected edges gives rank exactly $R+1$. The resulting subgraph has at most

$$(R+1)(2h+1) \le (4R-1)(2h+1) = W$$

edges, contrary to the assumption. □

Form a graph whose vertices are the simple cycles of G of length at most L, with two cycles adjacent when they share a vertex. The union of the cycles in one component is called a *cluster support*.

Lemma 7 (Size of a cluster support) *A cluster support S is connected, has minimum degree at least two, and satisfies*

$$1 \le r(S) \le R, \qquad |E(S)| \le r(S)L.$$

Distinct cluster supports have disjoint vertex sets.

Proof. Start with one cycle in the cluster. If the current union does not contain all cluster cycles, follow an intersection path to the first cycle with an edge outside the union, and add it. The new cycle meets the union. Its cycle-space vector is not in the old cycle space, because it contains a previously absent edge. Each addition therefore increases the rank by at least one.

If the rank first exceeded R, at most $R+1$ cycles would have been selected. Their connected union would have at most $(R+1)L \le W$ edges

and rank at least $R+1$, a contradiction. Since there are only finitely many cycles, the procedure ends with the whole support. If its final rank is r, at most r cycles were selected, proving the edge bound. A union of cycles has minimum degree at least two. Finally, two supports sharing a vertex would belong to the same component of the intersection graph. □

A *proper core* is a connected subgraph whose degrees are two or three and which has at least one degree-two vertex.

Lemma 8 (Disjoint buffers) *There is a possibly empty collection $\mathcal{K}$ of pairwise vertex-disjoint proper cores with the following properties:*

(i) Every simple cycle of length at most L lies in a member of $\mathcal{K}$.

(ii) Each $K \in \mathcal{K}$ has rank $r(K) \le R$ and at most $(2r(K)-1)L$ edges.

(iii) $N_h(K)$ consists of K and a tree attached at each degree-two vertex of K. Such a tree has 2^{j-1} vertices at depth j for $1 \le j \le h$, with no further edges inside $N_h(K)$ (Figure 1).

(iv) The vertex sets $V(N_h(K))$, for $K \in \mathcal{K}$, are pairwise disjoint.

Proof. Start with the cluster supports. Maintain vertex-disjoint connected subgraphs $\mathcal{K}_k$ at stage k such that, for each $K \in \mathcal{K}_k$,

$$1 \le r(K) \le R, \qquad 2 \le d_K(v) \le 3 \ (v \in V(K)), \qquad |E(K)| \le (2r(K)-1)L. \tag{9}$$

The disjointness invariant is $V(K) \cap V(K') = \varnothing$ for distinct $K, K' \in \mathcal{K}_k$. Lemma 7 supplies the initial bound, since $r \le 2r-1$ for $r \ge 1$. At intermediate stages we retain only the selected edges.

First perform a merger whenever two maintained subgraphs have distance at most $2h$. Choose a pair of minimum distance d and a shortest connecting path. The path has no internal vertex in either endpoint subgraph. It also has no internal vertex in a third maintained subgraph, since such an encounter would give a pair at distance less than d. Replace the pair and the connecting path by their union. The union has minimum degree at least two. If the old ranks were r_1, r_2, its rank is $r = r_1 + r_2$, because the path adds d edges and $d-1$ new vertices. More explicitly, with K^+ denoting the union,

$$\begin{aligned} |E(K^+)| &= |E(K_1)| + |E(K_2)| + d, \\ |V(K^+)| &= |V(K_1)| + |V(K_2)| + d - 1, \\ r(K^+) &= r(K_1) + r(K_2). \end{aligned}$$

Its edge count is at most

$$(2r_1 - 1)L + (2r_2 - 1)L + 2h \leq (2r - 1)L.$$

If $r > R$, then $r \leq 2R$ and the same subgraph has at most $(4R - 1)L = W$ edges, contrary to the assumption. Thus a merger preserves all the maintained conditions.

Only when all pairwise distances exceed $2h$ do we test for an expansion. Suppose that some K satisfies $r(N_h(K)) > r(K)$. Choose a breadth-first forest from the root set $V(K)$: every vertex outside K has one parent at smaller distance from K, and every tree has exactly one root in K. The union of K and this forest has rank $r(K)$, so $N_h(K)$ contains an additional edge. Add that edge and its two forest paths to K. At most $2h + 1 = L$ edges are added. The forest paths add the same number of edges as vertices, and the additional edge raises the rank by exactly one. If p is the number of newly retained forest edges, then

$$|V(K^+) \setminus V(K)| = p, \qquad |E(K^+) \setminus E(K)| = p + 1 \leq L,$$
$$r(K^+) = r(K) + 1.$$

Every new vertex has degree at least two in the enlarged subgraph. Indeed, before the additional edge is inserted, any leaf outside K in the union of the two paths must be an endpoint of that edge. All new vertices lie at distance at most h from the old K, so none lies in another maintained subgraph. If the old rank was r, the new edge count is at most

$$(2r - 1)L + L = 2rL \leq (2(r + 1) - 1)L.$$

An expansion from rank R would produce a forbidden subgraph with at most $2RL \leq W$ edges. Hence the new rank is still at most R. After each expansion, return to the merger test. An expansion can reduce the distance to another subgraph to at most $2h$; such new pairs are merged before any further expansion.

Let $F_k = \bigcup_{K \in \mathcal{K}_k} E(K)$. Every operation satisfies

$$F_k \subsetneq F_{k+1} \subseteq E(G), \qquad |F_{k+1}| \geq |F_k| + 1.$$

There are at most $|E(G)| - |F_0|$ operations, so the process terminates. At termination, the maintained subgraphs have pairwise distances greater than $2h$, and each satisfies $r(N_h(K)) = r(K)$. In the breadth-first forest just described, any edge of $N_h(K)$ outside the forest and K would raise the rank. There are no such edges. Thus $N_h(K)$ is exactly K with trees attached once.

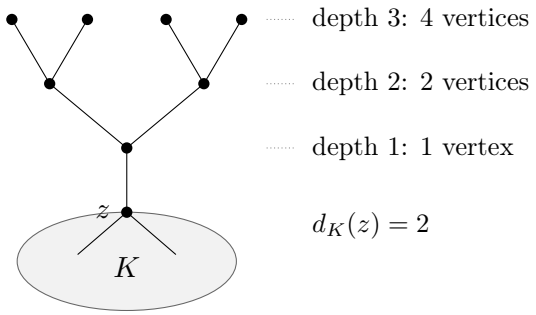


Figure 1: The first three levels of the tree attached at a degree-two vertex z of a core K. The shaded region represents the rest of the core. Only the two core edges incident with z are indicated; the other attached trees are omitted. Each vertex at depth $1 \le j < h$ has two children. At depth h, the two remaining ambient edges leave $N_h(K)$.

A vertex at depth less than h has all its ambient neighbours in the neighbourhood. Cubicity gives one child at each degree-two core vertex and two children at every subsequent nonterminal tree vertex. A degree-three core vertex has no neighbour outside the core. If a vertex w belonged to both $N_h(K)$ and $N_h(K')$, then

$$d_G(K, K') \le d_G(w, K) + d_G(w, K') \le 2h,$$

contradicting the stopping condition. Hence the neighbourhoods are vertex-disjoint. All initial short cycles have been retained.

Finally, a maintained subgraph with every degree equal to three would have no edge to its complement in G. Connectedness would force it to equal G, in which case $n = 2(r(G)-1) \le 2(R-1)$, contrary to the order hypothesis. Every final subgraph is therefore a proper core. □

4 A spectral cycle criterion

The next proposition isolates the hypotheses used by the spectral argument. Its parameters h and q may be chosen independently of the formulas in Theorem 1.

Proposition 9 (Spectral cycle criterion) *Let $R \ge 3$ be an integer, and let G be a connected simple cubic graph of order $n > 2(R-1)$. Set*

$$c = 16(R-1)^2, \qquad b = 2 - 1/c.$$

Let $h \geq 64c$ be an integer and let $q \geq 2h \log_2 n$ be an even integer. Put $L = 2h + 1$. Suppose that the following conditions hold.

(G1) *Every induced ball satisfies $r(B_G(v, h)) \leq R$.*

(G2) *There is a possibly empty collection $\mathcal{K}$ of proper cores $K \subseteq G$, each of rank at most R, such that every simple cycle of length at most L lies in a member of $\mathcal{K}$.*

(G3) *For each $K \in \mathcal{K}$, the graph*

$$\mathcal{F}_K = \big(V(N_h(K)),\ E(N_h(K)) \setminus E(K)\big)$$

is a forest, and each of its components contains exactly one vertex of K. For distinct $K, K' \in \mathcal{K}$,

$$V(N_h(K)) \cap V(N_h(K')) = \varnothing.$$

Define

$$a = (b/2)^h, \qquad \eta = 2n\, 2^{-q/2},$$

and

$$\begin{aligned} E = {} & 16c^2q^2a^2 + 64cqa^2 + (9c/4)q^2a \\ & + 12c^3q\, 2^{-h} + 3c(2R-1)\frac{q}{n}\, 2^{-h}. \end{aligned} \tag{10}$$

Let T_q and D_q count the cyclically nonbacktracking walks and the nonsimple ones, respectively, as in Section 2. Then

$$0 \leq \eta < \frac{1}{8}, \qquad T_q \geq (1-\eta)2^q > 0, \qquad \frac{D_q}{T_q} \leq \frac{E}{1-\eta}. \tag{11}$$

If $E < 1 - \eta$, the number c_q of undirected simple q-cycles satisfies

$$c_q \geq \frac{(1-\eta-E)2^q}{2q} > 0. \tag{12}$$

Through Section 8, retain the hypotheses and parameters of Proposition 9. The proof uses the following core and walk estimates and concludes in Section 8.4.

5 A resolvent bound for proper cores

Since $R \geq 3$, we have $c \geq 64$ and $b > \sqrt{3}$. For a proper core K with adjacency matrix A_K and degree matrix D_K, define

$$Q_\beta = \beta I + (D_K - I)/\beta - A_K.$$

For a real symmetric matrix M, the notation $M \succeq aI$ means $x^\top M x \geq a\|x\|^2$ for every real vector x.

Lemma 10 (Coercivity) *If $r(K) \leq R$, then $Q_\beta \succeq I/c$ for every $\beta \geq b$. The same bound holds with $+A_K$ in place of $-A_K$.*

Proof. Let Z and T be the degree-two and degree-three vertices of K. The degree sum gives

$$B_0 := |T| = 2(r(K) - 1) \leq 2(R - 1), \qquad Z \neq \varnothing.$$

At $\beta = 2$,

$$x^\top Q_2 x = \sum_{uv \in E(K)} (x_u - x_v)^2 + \frac{1}{2} \sum_{z \in Z} x_z^2.$$

For each $v \in T$, take a shortest path P_v to Z, ending at $z(v)$. It has at most B_0 edges, because all vertices before its endpoint are distinct members of T. Telescoping along the path, then applying Cauchy–Schwarz, yields

$$x_v^2 \leq 2x_{z(v)}^2 + 2B_0 \sum_{uw \in E(P_v)} (x_u - x_w)^2.$$

Summing over $v \in T$ gives

$$\begin{aligned} \|x\|^2 &\leq (2B_0 + 1) \sum_{z \in Z} x_z^2 + 2B_0^2 \sum_{uw \in E(K)} (x_u - x_w)^2 \\ &\leq 8(R-1)^2 x^\top Q_2 x. \end{aligned}$$

For the second inequality, use $2B_0^2 \leq 8(R-1)^2$ and $2B_0 + 1 \leq 4(R-1)^2$, valid for $R \geq 3$. The case $B_0 = 0$ is included. Consequently $Q_2 \succeq 2I/c$.

For $d \in \{2, 3\}$, the derivative of $\beta + (d-1)/\beta$ is $1 - (d-1)/\beta^2$. It is positive on $[b, \infty)$ and at most one on $[b, 2]$. Hence $0 \preceq Q_2 - Q_b \preceq I/c$, and $Q_b \succeq I/c$. Monotonicity gives the bound for all $\beta \geq b$.

For the sign variant, let $|x|$ be the coordinatewise absolute value. Term by term on the edges,

$$x^\top \big(\beta I + (D_K - I)/\beta + A_K\big) x \geq |x|^\top Q_\beta |x| \geq \|x\|^2/c.$$

□

Lemma 11 (Finite matrix resolvents) *Let M be a finite square matrix of positive order and let $\rho(M) = \max\{|\lambda| : \lambda \in \operatorname{spec}(M)\}$. For every real $s > \rho(M)$,*

$$(sI - M)^{-1} = \sum_{k \geq 0} s^{-k-1} M^k,$$

with absolute convergence in operator norm. If M is entrywise nonnegative, then $\rho(M)$ is an eigenvalue of M with a nonnegative unit eigenvector.

Proof. For a Jordan block $\lambda I + N$ of size d, where $N^d = 0$,

$$(\lambda I + N)^k = \sum_{\ell=0}^{\min\{k, d-1\}} \binom{k}{\ell} \lambda^{k-\ell} N^\ell.$$

For any $s_0 > \rho(M)$, the polynomial factors in k can be absorbed in a constant times s_0^k; blocks with $\lambda = 0$ vanish for $k \geq d$. Thus $\|M^k\| \leq C_{s_0} s_0^k$ for some finite constant C_{s_0}. Given $s > \rho(M)$, choose $\rho(M) < s_0 < s$. This proves absolute norm convergence. Multiplication of the partial sum by $sI - M$ gives $I - s^{-N-1} M^{N+1}$, which tends to I.

Now assume $M \geq 0$ entrywise. Choose a complex eigenvector z with $Mz = \lambda z$, $|\lambda| = \rho(M)$, and $\max_j |z_j| = 1$. With $\mathbf{1}$ denoting the all-ones vector, put

$$y_s = (sI - M)^{-1} \mathbf{1} = \sum_{k \geq 0} s^{-k-1} M^k \mathbf{1} \qquad (s > \rho(M)).$$

Entrywise nonnegativity gives $M^k \mathbf{1} \geq |M^k z| = \rho(M)^k |z|$, and hence

$$y_s \geq \frac{|z|}{s - \rho(M)}, \qquad \|y_s\| \longrightarrow \infty \quad \text{as } s \downarrow \rho(M).$$

The nonnegative unit vectors $x_s = y_s / \|y_s\|$ have a convergent subsequence as $s \downarrow \rho(M)$. Its limit x has norm one and satisfies

$$(sI - M) x_s = \frac{\mathbf{1}}{\|y_s\|} \longrightarrow 0, \qquad Mx = \rho(M) x.$$

□

Lemma 12 (The walk generating function) *Let K be a proper core of rank at most R. Let W_j count its internally nonbacktracking walks of length j, with $W_0 = I$. For $b \leq \beta \leq 2$, the series*

$$F_K(1/\beta) = \sum_{j \geq 0} \beta^{-j} W_j = (\beta - \beta^{-1}) Q_\beta^{-1}$$

converges in operator norm and has norm at most $3c/2$. *Consequently, for every vector* x *on* $V(K)$,

$$\sum_{C \text{ simple in } K} \beta^{-|C|}\|x_C\|^2 \le \frac{3c}{4}\|x\|^2. \tag{13}$$

Proof. Precisely, $W_j(v,w) = |\mathcal{W}_j^K(v,w)|$. Reversing a walk gives $W_j^\top = W_j$, and every entry of W_j is nonnegative. We have $W_1 = A_K$, $W_2 = A_K^2 - D_K$, and

$$W_j = A_K W_{j-1} - (D_K - I)W_{j-2} \qquad (j \ge 3).$$

To justify the last identity, prepend an edge at a vertex v. For each remaining walk of length $j-2$, exactly $d_K(v)-1$ choices produce a forbidden reversal. If $F(z) = \sum_{j\ge 0} z^j W_j$, the coefficient calculation is

$$\begin{aligned}[I - zA_K + z^2(D_K - I)]F(z) &= I + z(W_1 - A_K) \\ &\quad + z^2[W_2 - A_K W_1 + D_K - I] \\ &\quad + \sum_{j\ge 3} z^j[W_j - A_K W_{j-1} + (D_K - I)W_{j-2}] \\ &= (1-z^2)I.\end{aligned}$$

These are formal power-series identities; they also hold analytically near zero. Therefore

$$F(z) = (1-z^2)[I - zA_K + z^2(D_K - I)]^{-1}. \tag{14}$$

The corresponding regular-graph identity appears in [Ran18, Section 2.2]; the computation above keeps the degree matrix D_K in its original order.

Let B_K be the directed-edge nonbacktracking matrix of K. Its row sums are at most two. If its spectral radius ρ were at least $b > 1$, Lemma 11 would give the real eigenvalue ρ. Since $1 - \rho^{-2} \ne 0$, Lemma 2 would imply

$$0 = \det(I - \rho^{-1}B_K) \quad \Longrightarrow \quad \det\bigl(I - \rho^{-1}A_K + \rho^{-2}(D_K - I)\bigr) = 0.$$

The matrix on the right is Q_ρ/ρ, contradicting $Q_\rho \succeq I/c$. Therefore

$$\rho(B_K) < b. \tag{15}$$

For $j \ge 1$, W_j is obtained from B_K^{j-1} by multiplication by the tail and head incidence matrices. More explicitly, $W_j = UB_K^{j-1}V^\top$, with U, V as in the proof of Lemma 2. Lemma 11 and (15) give

$$\sum_{j\ge 1} \beta^{-j} W_j = \beta^{-1}U(I - \beta^{-1}B_K)^{-1}V^\top \qquad (\beta \ge b),$$

with absolute convergence in operator norm. The recurrence can therefore be summed at $z = 1/\beta$. Since

$$I - \beta^{-1}A_K + \beta^{-2}(D_K - I) = \beta^{-1}Q_\beta,$$

formula (14) gives $F_K(1/\beta) = (\beta - \beta^{-1})Q_\beta^{-1}$. For $b \le \beta \le 2$,

$$\|F_K(1/\beta)\| \le (\beta - \beta^{-1})\|Q_\beta^{-1}\| \le \frac{3}{2}c.$$

Each simple cycle through v contributes two orientations to the v, v entry of the series, and all coefficients are nonnegative. It follows that

$$2\sum_{C \ni v} \beta^{-|C|} \le F_K(1/\beta)_{vv} - 1 \le 3c/2.$$

Multiply by $|x_v|^2$ and sum over v to obtain (13). □

6 Eigenvector mass on short cycles

Lemma 13 (Mass in a tree buffer) *Let $A_G\phi = \lambda\phi$, where $|\lambda| > 2\sqrt{2}$, and put $\beta = (|\lambda| + \sqrt{\lambda^2 - 8})/2$. If $\beta \ge b$ and $K \in \mathcal{K}$, with $N = N_h(K)$, then*

$$\|\phi_K\|^2 \le c^2(2/\beta^2)^h\|\phi_N\|^2.$$

Proof. First suppose $\lambda > 0$. Define $a_0 = 0$, $a_1 = 1$, and $a_{j+1} = \lambda a_j - 2a_{j-1}$. With $\gamma = 2/\beta < \beta$,

$$a_j = \frac{\beta^j - \gamma^j}{\beta - \gamma} = \sum_{k=0}^{j-1} \beta^{j-1-k}\gamma^k \qquad (j \ge 1).$$

Thus $a_j \ge \beta^{j-1}$ and $a_{j-1}/a_j \le 1/\beta$. The latter also follows from $a_j = \beta a_{j-1} + \gamma^{j-1}$.

Write $x = \phi_K$ and let Z_K be the degree-two vertices of K. For $v \in Z_K$, denote the depth-j shell of its attached tree by $\Gamma_j(v)$. By hypothesis (G3), the components outside K are trees with a single attachment to K. A degree-two core vertex has one child, and a vertex at depth $1 \le j < h$ has two children: all its ambient neighbours lie in $N_h(K)$, and exactly one is its parent. The forest condition prevents identifications or additional edges. Thus

$$|\Gamma_j(v)| = 2^{j-1} \quad (1 \le j \le h), \qquad V(N) = V(K) \,\dot\cup \coprod_{v \in Z_K} \coprod_{j=1}^{h} \Gamma_j(v).$$

Define the vector $S_j \in \mathbb{R}^{V(K)}$ by

$$S_j(v) = \begin{cases} \displaystyle\sum_{w \in \Gamma_j(v)} \phi_w, & v \in Z_K, \\ 0, & d_K(v) = 3. \end{cases}$$

The eigenvector equations at the core and at depth one give

$$S_1 = (\lambda I - A_K)x, \qquad S_2 = \lambda S_1 - (3I - D_K)x.$$

At subsequent depths, summing the eigenvector equations gives $S_{j+1} = \lambda S_j - 2S_{j-1}$ for $2 \le j < h$. The factor two counts the two children of each parent in the preceding shell. Induction yields, for $1 \le j \le h$,

$$S_j = a_j M_j x, \qquad M_j = \lambda I - A_K - \frac{a_{j-1}}{a_j}(3I - D_K).$$

Since $3I - D_K$ is nonnegative diagonal,

$$M_j \succeq \lambda I - A_K - (3I - D_K)/\beta = Q_\beta \succeq I/c.$$

For a symmetric matrix satisfying this bound, $\|M_j x\| \ge \|x\|/c$, by Cauchy–Schwarz applied to $x^\top M_j x$. Therefore $\|S_j\| \ge \beta^{j-1}\|x\|/c$.

Each nonzero coordinate of S_j sums 2^{j-1} vertex values. All these shells are disjoint. Cauchy–Schwarz gives

$$\begin{aligned} \|\phi_N\|^2 &\ge \|x\|^2 + \sum_{j=1}^{h} 2^{-(j-1)} \|S_j\|^2 \\ &\ge \left(1 + c^{-2} \sum_{j=0}^{h-1} (\beta^2/2)^j\right) \|x\|^2 \\ &\ge c^{-2}(\beta^2/2)^h \|x\|^2. \end{aligned}$$

For the last step, set $z = \beta^2/2 \in [1, 2]$ and use

$$z^h = 1 + (z-1)\sum_{j=0}^{h-1} z^j \le c^2 + \sum_{j=0}^{h-1} z^j.$$

If $\lambda < 0$, put $\widehat{\lambda} = |\lambda|$ and $\widehat{S}_j = (-1)^j S_j$. Direct substitution gives

$$\begin{aligned} \widehat{S}_1 &= (\widehat{\lambda} I + A_K)x, \\ \widehat{S}_2 &= \widehat{\lambda}\widehat{S}_1 - (3I - D_K)x, \\ \widehat{S}_{j+1} &= \widehat{\lambda}\widehat{S}_j - 2\widehat{S}_{j-1} \quad (2 \le j < h). \end{aligned}$$

Use the same positive sequence a_j, now with parameter $\widehat{\lambda}$. The corresponding matrix is

$$\widehat{M}_j = \widehat{\lambda} I + A_K - \frac{a_{j-1}}{a_j}(3I - D_K) \succeq \beta I + (D_K - I)/\beta + A_K \succeq I/c.$$

Since $\|\widehat{S}_j\| = \|S_j\|$, every shell-norm estimate is unchanged. □

Let $\mathcal{C}_h$ be the set of simple cycles of length at most $L = 2h + 1$.

Lemma 14 (Two short-cycle sums) *For an eigenvector and parameter as in Lemma 13,*

$$\sum_{C \in \mathcal{C}_h} \beta^{-|C|} \|\phi_C\|^2 \le \frac{3c^3}{4}(2/\beta^2)^h \|\phi\|^2. \tag{16}$$

For every $b \le \beta \le 2$,

$$\sum_{C \in \mathcal{C}_h} |C| \beta^{-|C|} \le \frac{3c}{4}(2R-1)n\, 2^{-h}. \tag{17}$$

Proof. Each short cycle lies in a unique member of $\mathcal{K}$. Apply (13) in each core, followed by Lemma 13. The neighbourhoods are vertex-disjoint, so $\sum_K \|\phi_{N_h(K)}\|^2 \le \|\phi\|^2$. This proves (16).

For the second bound, let Z_K be the degree-two vertices of K. There are at most $2R-2$ other vertices and $|Z_K| \ge 1$. Hence

$$|V(K)| \le (2R-1)|Z_K|.$$

The buffer has $|V(K)| + |Z_K|(2^h - 1) \ge |Z_K| 2^h$ vertices. Disjointness gives

$$\sum_K |V(K)| \le (2R-1)2^{-h} \sum_K |V(N_h(K))| \le (2R-1)n\, 2^{-h}.$$

Apply (13) to the all-ones vector in each core and sum. Its restriction to C has squared norm $|C|$. □

7 Local walk counts

Lemma 15 (Entry and second-moment bounds) *For all vertices v, w,*

$$(P_h(A_G))_{vw} \le (3c/2)b^h, \tag{18}$$

$$(P_h(A_G)^2)_{vv} \le (9c^2/4 + 3/2)b^{2h}. \tag{19}$$

Consequently, with $a = (b/2)^h$, every $j \ge h+1$ satisfies

$$f_j(v) \le (9c/4)2^j a. \tag{20}$$

Proof. Fix v and let $J = B_G(v, h)$. Every length-h walk from v lies in J, and hypothesis (G1) gives $r(J) \le R$. The 2-core of J is obtained by repeatedly deleting vertices of degree at most one. If nonempty, it is connected: deleting a leaf from a connected graph does not disconnect the remaining vertices. It has minimum degree two, maximum degree three, and rank $r(J)$. It cannot be cubic, since a cubic subgraph of connected G must be all of G, whose rank is $n/2 + 1 > R$. Thus it is a proper core; denote it by K.

Each component outside K is a tree attached to K by one edge. To see this, a cycle outside K, or a path outside K joining two core vertices, could not be deleted by the leaf-pruning procedure. Connectedness of J ensures at least one attachment. Equivalently, deleting the core edges from J leaves a forest whose components each contain exactly one core vertex. If K is empty, J is a tree and only the tree-path counts below are needed.

Suppose K is nonempty. A nonbacktracking walk cannot leave K and return through the same attached tree, since a nonempty closed walk in a tree contains an immediate reversal. A walk that uses a core edge therefore consists of the unique tree path from v to its core entry x, a walk within K, and the unique tree path from its exit y to its endpoint. Let a_0 be the length of the initial tree path. For a fixed endpoint w, its exit vertex y and final depth d are fixed, so the number of these walks is at most $W_{h-a_0-d}(x, y)$, interpreted as zero for negative indices. Lemma 12 gives

$$W_r(x, y) \le (3c/2)b^r \le (3c/2)b^h \qquad (0 \le r \le h).$$

A walk using no core edge is a simple path in the forest and has at most one realization for a fixed endpoint.

These two types cannot reach the same endpoint at the same total length. If the tree-only path exists, v and w lie in the same rooted tree component. A walk using core edges then makes a nonempty return from that root through the core, in addition to its two tree paths. Its length is strictly greater than the tree distance between v and w. This proves (18) and also rules out cross terms between the two types in the squared row norm.

For a fixed exit y, there are at most 2^d possible endpoints at depth d, including $d = 0$. This follows from subcubicity and the fact that a core vertex has at most one neighbour outside the core. Since $b^2 > 2$, the squared

contribution of the core-using walks is at most

$$\sum_{\substack{d\ge 0,\ y\in V(K)\\ h-a_0-d\ge 0}} 2^d W_{h-a_0-d}(x,y)^2 \le b^{2h} \sum_{r\ge 0,\ y\in V(K)} b^{-2r} W_r(x,y)^2$$
$$\le b^{2h} \sum_{y\in V(K)} \left(\sum_{r\ge 0} b^{-r} W_r(x,y)\right)^2$$
$$\le (9c^2/4) b^{2h}.$$

In the first inequality, use $2^d \le b^{2d} \le b^{2(h-r)}$ when $r = h - a_0 - d$. The second uses nonnegativity, and the last is the squared row-norm bound from Lemma 12.

Tree-only paths have distinct endpoints and contribute at most the total number $3 \cdot 2^{h-1}$ of length-h nonbacktracking walks from v in G. This is at most $(3/2)b^{2h}$. Adding the two contributions proves (19), since $P_h(A_G)$ is symmetric.

Finally, split a return of length $j \ge h+1$ after its first $j-h$ edges. There are $3 \cdot 2^{j-h-1}$ possible prefixes. For each one, (18) bounds the number of length-h suffixes returning to v. Ignoring the transition condition at the split gives

$$f_j(v) \le 3 \cdot 2^{j-h-1}(3c/2)b^h = (9c/4)2^j (b/2)^h.$$

□

8 Bounding nonsimple closed walks

Set

$$t = \log_2 n, \qquad \delta = 2t/q, \qquad \beta_* = 2^{1-\delta}.$$

The parameter hypotheses in Proposition 9 give

$$c \ge 64, \quad t \ge 2, \quad h \ge 64c, \quad q \ge 2ht \ge 4h. \tag{21}$$

In particular,

$$\delta h \le 1, \qquad n(\beta_*/2)^{q/2} = 1, \qquad n(\beta_*/2)^q = 1/n. \tag{22}$$

Also

$$\eta \le 2n^{1-h} < 1/8.$$

Using $e^{-x} \geq 1-x$ and $\delta \leq 1/(64c)$ gives $\beta_* \geq 2 - 2\delta \ln 2$ and $\beta_* - b \geq 1/(2c)$. Thus $b < \beta_* < 2$ and

$$\frac{1}{1 - b/\beta_*} \leq 4c, \tag{23}$$

$$(b/\beta_*)^{2h} = a^2 2^{2\delta h} \leq 4a^2, \tag{24}$$

$$(2/\beta_*^2)^h = 2^{-h} 2^{2\delta h} \leq 4 \cdot 2^{-h}. \tag{25}$$

Choose a real orthonormal eigenbasis $\phi_1, \ldots, \phi_n$ for A_G. Write $\phi_{i,C}$ for the restriction of ϕ_i to $V(C)$, and put $w_i(v) = \phi_i(v)^2$. Orthonormality and normalization give

$$w_i(v) \geq 0, \qquad \sum_{i=1}^{n} w_i(v) = 1, \qquad \sum_{v \in V(G)} w_i(v) = 1. \tag{26}$$

For every integer $j \geq 0$, the spectral theorem gives

$$\begin{aligned} f_j(v) &= \sum_{i=1}^{n} w_i(v) P_j(\lambda_i), \\ (P_j(A_G)^2)_{vv} &= \sum_{i=1}^{n} w_i(v) P_j(\lambda_i)^2. \end{aligned} \tag{27}$$

Write

$$H_j(v) = \sum_{\beta_i > b} w_i(v) \beta_i^j,$$

$$H_j^*(v) = \sum_{\beta_i > \beta_*} w_i(v) \beta_i^j, \qquad\qquad S_q^* = \sum_{\beta_i > \beta_*} \beta_i^q.$$

Since $b^2 > 3$, both C_b and C_{β_*} are at most two. Lemma 4 and the spectral decomposition give, for $j \geq 1$,

$$f_j(v) \leq 2H_j(v) + 2b^j, \qquad f_j(v) \leq 2H_j^*(v) + 2\beta_*^j. \tag{28}$$

For either $\alpha = b$ or $\alpha = \beta_*$, Lemma 4 gives explicitly

$$\begin{aligned} f_j(v) &\leq \sum_{i \in \mathcal{I}_{>\alpha}} w_i(v) |P_j(\lambda_i)| + \sum_{i \notin \mathcal{I}_{>\alpha}} w_i(v) |P_j(\lambda_i)| \\ &\leq 2 \sum_{i \in \mathcal{I}_{>\alpha}} w_i(v) \beta_i^j + 2\alpha^j \sum_{i \notin \mathcal{I}_{>\alpha}} w_i(v) \\ &\leq 2 \sum_{i \in \mathcal{I}_{>\alpha}} w_i(v) \beta_i^j + 2\alpha^j. \end{aligned}$$

The index complement includes the complex-root spectral interval and all real-root parameters at most α.

8.1 A weighted correlation bound

The second identity in (27) and the lower polynomial bound give

$$\frac{9}{4}\sum_{\beta_i>b} w_i(v)\beta_i^{2h} \le \sum_{\beta_i>b} w_i(v)P_h(\lambda_i)^2 \le (P_h(A_G)^2)_{vv}.$$

Combining this with (19) yields

$$\sum_{\beta_i>b} w_i(v)\beta_i^{2h} \le \frac{4}{9}(P_h(A_G)^2)_{vv} \le (c^2+2/3)b^{2h}. \tag{29}$$

For $r+s=q$ with $r,s\ge 2h$, we claim that

$$\sum_v H_r(v)H_s^*(v) \le (c^2+2/3)(b/\beta_*)^{2h}S_q^*. \tag{30}$$

For nonnegative weights t_i and two nondecreasing functions f,g on the same ordered parameters,

$$\left(\sum_i t_i f_i\right)\left(\sum_i t_i g_i\right) \le \left(\sum_i t_i\right)\left(\sum_i t_i f_i g_i\right).$$

Subtracting the left side from the right side gives $\frac{1}{2}\sum_{i,j} t_i t_j (f_i-f_j)(g_i-g_j) \ge 0$. Apply this at a fixed vertex with

$$t_i = w_i(v)\beta_i^{2h}, \quad f_i = \beta_i^{r-2h}, \quad g_i = \beta_i^{s-2h}\mathbf{1}_{\{\beta_i>\beta_*\}},$$

where $\beta_i > b$. Both functions are nondecreasing. By (29),

$$\begin{aligned} H_r(v)H_s^*(v) &\le (c^2+2/3)b^{2h}\sum_{\beta_i>\beta_*} w_i(v)\beta_i^{q-2h} \\ &\le (c^2+2/3)(b/\beta_*)^{2h}\sum_{\beta_i>\beta_*} w_i(v)\beta_i^{q}. \end{aligned}$$

Summing over v proves (30).

8.2 Short marked cycles

Define

$$D_{\text{short}} = q \sum_{r=3}^{\min\{L,\lfloor q/2\rfloor\}} \sum_{v} s_r(v) f_{q-r}(v),$$

$$D_{\text{long}} = q \sum_{r=L+1}^{\lfloor q/2\rfloor} \sum_{v} s_r(v) f_{q-r}(v).$$

An empty sum is zero. Lemma 5 gives $D_q \le D_{\text{short}} + D_{\text{long}}$.

Each undirected simple cycle contributes two orientations at every one of its vertices. Applying the second bound in (28), and enlarging the cycle sum to all of $\mathcal{C}_h$, gives

$$\begin{aligned} D_{\text{short}} \le 4q \sum_{\beta_i > \beta_*} \beta_i^q \sum_{C \in \mathcal{C}_h} \beta_i^{-|C|} \|\phi_{i,C}\|^2 \\ + 4q\beta_*^q \sum_{C \in \mathcal{C}_h} |C| \beta_*^{-|C|}. \end{aligned}$$

All complementary lengths are positive, since $q \ge 4h$. Use (16) on each normalized eigenvector, then (25); use (17) for the second sum. The result is

$$D_{\text{short}} \le 12c^3 q\, 2^{-h} S_q^* + 3c(2R-1)nq\, 2^{-h} \beta_*^q. \tag{31}$$

8.3 Long marked cycles

For $2h+2 \le r \le \lfloor q/2 \rfloor$, put $s = q - r$ and

$$G_r = (9c/4)a\, 2^r, \qquad L_r = \min\{G_r, 2b^r\}.$$

Equations (20) and (28) give $f_r \le G_r$ and $f_r \le 2H_r + L_r$. For the latter inequality, use $f_r \le G_r \le 2H_r + G_r$ when $G_r \le 2b^r$, and the spectral bound when $2b^r < G_r$. Thus, for every $v \in V(G)$,

$$\begin{aligned} f_r(v) f_s(v) &\le 2 f_r(v) H_s^*(v) + 2 f_r(v) \beta_*^s \\ &\le 2[2H_r(v) + L_r] H_s^*(v) + 2G_r \beta_*^s \\ &= 4H_r(v) H_s^*(v) + 2L_r H_s^*(v) + 2G_r \beta_*^s. \end{aligned}$$

There are at most $q/2$ possible values of r, and $r, s \ge 2h$. For the first term, (30) and (24) give

$$\begin{aligned} 4q \sum_r \sum_v H_r(v) H_s^*(v) &\le 8(c^2 + 2/3) q^2 a^2 S_q^* \\ &\le 16 c^2 q^2 a^2 S_q^*. \end{aligned}$$

For the second, $\sum_v H_s^*(v) \le \beta_*^{-r} S_q^*$ and

$$\sum_{r \ge 2h+2} L_r \beta_*^{-r} \le \frac{2(b/\beta_*)^{2h}}{1 - b/\beta_*} \le 32ca^2.$$

Its total contribution is consequently at most $64cqa^2 S_q^*$. For the last term, $s \ge q/2$ and (22) imply

$$n 2^r \beta_*^s = 2^q n (\beta_*/2)^s \le 2^q.$$

Including the q mark positions and at most $q/2$ lengths gives

$$2q \sum_r \sum_v G_r \beta_*^s \le (9c/4) q^2 a \, 2^q.$$

Since $s_r(v) \le f_r(v)$, these bounds prove

$$D_{\text{long}} \le (16c^2 q^2 a^2 + 64cqa^2) S_q^* + (9c/4) q^2 a \, 2^q. \tag{32}$$

An empty range of long lengths contributes zero and satisfies the same bound.

8.4 The normalized error

Proof. We now complete the proof of Proposition 9. The bound $\eta < 1/8$ follows from (21). Lemma 3 then gives

$$T_q \ge (1 - \eta) 2^q > 0, \qquad S_q^* \le \frac{T_q}{1 - \eta}, \qquad 2^q \le \frac{T_q}{1 - \eta}.$$

Set

$$A_0 = 16c^2 q^2 a^2 + 64cqa^2 + 12c^3 q \, 2^{-h},$$
$$B_0 = (9c/4) q^2 a, \qquad C_0 = 3c(2R - 1) nq \, 2^{-h}.$$

Equations (31) and (32) say $D_q \le A_0 S_q^* + B_0 2^q + C_0 \beta_*^q$. Division by $T_q > 0$ therefore gives

$$\frac{D_q}{T_q} \le \frac{A_0 + B_0 + C_0 (\beta_*/2)^q}{1 - \eta}.$$

The third coefficient simplifies using

$$nq(\beta_*/2)^q = nq \, 2^{-\delta q} = nq \, 2^{-2t} = q/n,$$

which gives

$$\frac{D_q}{T_q} \le \frac{E}{1-\eta} \tag{33}$$

with E as in (10). If $E < 1-\eta$, then $1 - E/(1-\eta) > 0$. Applying (2) and the lower bound for T_q gives

$$\begin{aligned} 2qc_q = T_q - D_q &\ge \left(1 - \frac{E}{1-\eta}\right) T_q \\ &\ge \left(1 - \frac{E}{1-\eta}\right)(1-\eta)2^q \\ &= (1-\eta-E)2^q > 0. \end{aligned}$$

This proves (12) and completes the proposition. □

9 Parameter estimates and proof of the theorem

Return to the parameters of Theorem 1; in particular, $t = \log_2 n$, $u = \log_2(t+2)$, and $q = \mathrm{pow}_2(2ht)$. They satisfy

$$c \ge 64, \qquad t \ge 2, \qquad u \ge 2, \qquad h \ge 64c, \qquad 2ht \le q < 4ht.$$

Indeed, the bracket defining h is at least $2+2+2+10 = 16$. Also q is a power of two with $q \ge 4h \ge 4$, so it is even. These are the parameter hypotheses of Proposition 9. Let a, η, and E have the values specified there.

The definition of h and $1 - x \le e^{-x} \le 2^{-x}$ for $0 \le x < 1$ imply

$$a \le 2^{-h/(2c)} \le \frac{2^{-20}}{(t+2)^2(u+2)^2(c+1)^4} \le \frac{1}{64c}. \tag{34}$$

Also, $b \ge \sqrt{2}$ and $h \ge 64c$ give

$$\frac{2^{-h}}{a} = b^{-h} \le 2^{-h/2} \le c^{-4}. \tag{35}$$

The last step follows from $h/2 \ge 32c \ge 4\log_2 c$.

Divide (10) by cq^2a and use these two bounds:

$$\begin{aligned} \frac{E}{cq^2a} &= 16ca + \frac{64a}{q} + \frac{9}{4} + \frac{12c^2}{q}\frac{2^{-h}}{a} + \frac{3(2R-1)}{nq}\frac{2^{-h}}{a} \\ &\le \frac{1}{4} + \frac{1}{cq} + \frac{9}{4} + \frac{12}{qc^2} + \frac{6R}{nqc^4} < 4. \end{aligned}$$

Indeed, $R \le c$, $n \ge 4$, and $q \ge 4h \ge 256c$, so each of the second, fourth, and fifth displayed terms is less than $1/4$. Thus

$$E < 4cq^2a. \tag{36}$$

For $u \ge 2$ and $c \ge 64$,

$$\log_2(u+2) \le u, \qquad \log_2(c+1) \le \sqrt{c}.$$

For the first inequality, the difference is zero at $u = 2$ and has positive derivative $1 - 1/((u+2)\ln 2)$. For the second, the difference is positive at $c = 64$ and has positive derivative $1/(2\sqrt{c}) - 1/((c+1)\ln 2)$ on $[64, \infty)$. Including the ceiling in the definition of h, we have

$$\begin{aligned} h &\le 4c(2u + 2\sqrt{c} + 10) + 1 \\ &\le 4c^{3/2}(u+2). \end{aligned}$$

The second line is equivalent to $(\sqrt{c} - 2)u \ge 10 + 1/(4c)$; its left side is at least 12. Since $q < 4ht$, equations (34) and (36) now yield

$$\begin{aligned} E &< 64ch^2t^2a \\ &\le 1024c^4(u+2)^2t^2a \\ &\le \frac{1}{1024}\left(\frac{c}{c+1}\right)^4\left(\frac{t}{t+2}\right)^2 < \frac{1}{1024}. \end{aligned}$$

Proof. To prove Theorem 1, suppose that its rank-witness alternative fails. Lemma 6 verifies hypothesis (G1) of Proposition 9. Lemma 8 supplies the proper cores and disjoint tree neighbourhoods required by hypotheses (G2) and (G3). The order restriction is part of the theorem, and the parameter conditions were checked above. All hypotheses of the proposition therefore hold.

The estimates $E < 1/1024$ and $\eta < 1/8$ give

$$\frac{D_q}{T_q} \le \frac{E}{1-\eta} < \frac{1/1024}{7/8} = \frac{1}{896} < 1.$$

The cycle-count bound in the proposition gives

$$c_q \ge \frac{(1 - \eta - E)2^q}{2q} > 0.$$

Thus G contains a simple cycle of length q.

Finally, $c = O(R^2)$, $\log_2(c+1) = O(\log R)$, and $\log_2(u+2) \le u$ imply $h = O(R^2(u + \log R))$. Since $t \ge 2$, $u = \log_2(t+2) = O(\log\log n)$. Substitution into $W = (4R-1)(2h+1)$ proves the asserted order bound. $\square$

10 Growing rank and weighted cubic kernels

For the multigraphs in this section, a loop contributes two to the degree. A cycle is a connected 2-regular subgraph; thus a loop and a pair of parallel edges are allowed cycles. The *weighted length* of a cycle is the sum of its edge lengths.

Lemma 16 (Kernel extraction) *Let J be a connected subcubic graph of rank $R+1$, where $R \geq 1$. Pruning leaves and suppressing maximal degree-two chains produces a connected cubic multigraph with $2R$ vertices and $3R$ edges. Each kernel edge has a positive integral length, and the sum of these lengths is at most $|E(J)|$. Simple cycles of J correspond bijectively to cycles of the kernel, with lengths preserved as weighted lengths.*

Proof. Deleting a leaf and its incident edge preserves connectedness of the remaining graph and preserves its rank. Since the rank is at least two, this process leaves a nonempty connected 2-core. If its degree-two and degree-three vertex counts are n_2, n_3, then

$$R+1 = \frac{2n_2+3n_3}{2} - (n_2+n_3) + 1 = \frac{n_3}{2} + 1.$$

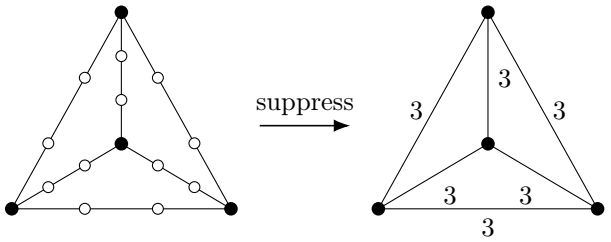
Hence $n_3 = 2R > 0$.

Follow an edge from a degree-three vertex through successive degree-two vertices until reaching a degree-three vertex, possibly the starting vertex. Such a chain cannot become trapped in a separate cycle of degree-two vertices: that cycle would be a component of the connected core and would leave no degree-three vertices. These maximal chains partition the core edges and have disjoint interiors. Replace each by one edge, retaining its number of edges as its positive integral length. The resulting multigraph is connected and cubic. It has $2R$ vertices, so its degree sum gives $3R$ edges. The retained lengths sum to the number of core edges, at most $|E(J)|$.

A simple cycle cannot contain a pruned leaf edge. At a degree-two vertex of the core it uses both incident edges, so it traverses each maximal chain in full. Suppression therefore takes it to a connected 2-regular kernel subgraph. Conversely, expanding such a kernel cycle along the corresponding internally disjoint chains gives a simple cycle of J. The two operations are inverse, and expansion adds exactly the retained edge lengths. □

Remark 17 (Subdivision lengths) *Replace every edge of K_4 by a chain of three edges, as in Figure 2. The resulting simple subcubic graph H has* 16 *vertices,* 18 *edges, and rank* 3*. Its twelve subdivision vertices have degree*

two. The cycles of K_4 consist of four triangles and three 4-cycles: a triangle is determined by its three vertices, and a 4-cycle by a cyclic ordering of the four vertices up to rotation and reversal, giving $4!/8 = 3$ choices. Every cycle of H traverses each subdivision chain in full. Thus H has four cycles of length 9 and three of length 12, and is dyadic-free. Its cubic kernel is K_4 with length three on every edge. These lengths retain the cycle lengths 9 and 12 under suppression; the underlying unweighted kernel has 4-cycles.



H: subdivided K_4
16 vertices, 18 edges
Rank 3

Cubic kernel K_4
6 edges of length 3
Rank 3

Figure 2: Suppression with lengths retained. Filled vertices have degree three; open vertices have degree two. A kernel triangle expands to a 9-cycle, and a kernel 4-cycle expands to a 12-cycle. Both graphs have cycle rank 3.

Corollary 18 (A logarithmic-size witness) *Let G be a connected simple dyadic-free cubic graph of order n. Set*

$$t = \log_2 n, \qquad u = \log_2(t+2), \qquad R = \left\lfloor \left(\frac{t}{4096u} \right)^{1/3} \right\rfloor.$$

If $u \geq 20$ and $R \geq 3$, then G contains a connected subgraph J satisfying

$$|E(J)| < t, \qquad r(J) = R + 1 > \frac{1}{16}(t/u)^{1/3}.$$

Its pruned and suppressed kernel has $2R$ vertices and $3R$ edges, with positive integral edge lengths of total less than t and with no cycle of dyadic weighted length.

Proof. We have $R \leq t^{1/3}$ and $n = 2^t > 2t^{1/3}$ in the stated range, so $n > 2(R-1)$. Theorem 1 applies. Its cycle alternative is excluded because

q is a power of two greater than four. Thus there is a witness with at most $W = (4R-1)(2h+1)$ edges.

Now $c+1 \le 32R^2$, and

$$\log_2 R \le \tfrac{1}{3}\log_2 t \le u/3.$$

Also $\log_2(u+2) \le u/4$ for $u \ge 20$: at 20 the inequality is $\log_2 22 < 5$, and the difference has positive derivative $1/4 - 1/((u+2)\ln 2)$ thereafter. Consequently the bracket in the definition of h is at most

$$u + u/4 + 2(5 + 2u/3) + 10 = 31u/12 + 20 \le 4u.$$

The last inequality follows from $u \ge 20$. Since $c \le 16R^2$,

$$h \le 16cu + 1 \le 256R^2u + 1 \le 257R^2u.$$

It follows that

$$\begin{aligned} W &< 8Rh + 4R \\ &\le 2056R^3u + 4R \\ &\le 2057R^3u \le \frac{2057}{4096}t < t. \end{aligned}$$

Here $4R \le R^3u$ because $R \ge 3$ and $u \ge 20$. Finally, the strict inequality $\lfloor x \rfloor + 1 > x$ gives

$$R + 1 > \left(\frac{t}{4096u}\right)^{1/3} = \frac{1}{16}(t/u)^{1/3}.$$

Lemma 16 supplies the kernel and preserves every cycle length. Since its expanded cycles lie in the dyadic-free graph G, none has dyadic weighted length. □

In particular, the corollary gives a witness with $O(\log n)$ edges and rank $\Omega((\log n/\log\log n)^{1/3})$ for every sufficiently large connected dyadic-free cubic graph.

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